

Good morning, good afternoon, good evening. Thank you for taking the time to watch my presentation. My name is Frank Ademoye and today I'd like to present ResearchMate, an intelligent academic research planning and decision support agent that I designed and implemented as my individual development project. So the idea behind research mate came from a simple observation. Throughout my studies, I noticed that many students know what they want to research, but struggle to transform those ideas into structured academic research plans. So, my objective simply wasn't to build another AI application. Instead, I wanted to design an intelligent agent that receives a research goal, plans the activities required to investigate that goal, processes the information, and produces a structured academic research plan. So during this presentation, I'll explain the problem I set out to solve, the architecture I designed, how the intelligent agent works, how I tested it, and finally evaluate both its strengths, limitations before discussing future plans and developments.

So before writing any code, I wanted to understand the real problem I was trying to solve. One issue I repeatedly identified was that students often begin with broad research ideas but struggle to convert those ideas into structured academic research plans. They may find it difficult to define a research aim, align research objectives, formulate meaningful research questions or even know where to begin searching the academic literature. So although modern AI tools can generate useful responses, they generally don't answer whatever prompt is they generally answer, excuse me, whatever prompt is provided. They don't necessarily guide users through the planning process itself. That observation became the motivation for researchmate. So I designed a researchmate to support users before they begin writing by helping them organise their thinking into a structured research plan. More importantly the goal is not to replace academic judgment but to support it.

So once the problem so once I understood the problem, excuse me I defined a clear aim for the project. The aim was to design and implement an intelligent planning agent capable of transforming a user's research topic into a structured academic research plan. To achieve that I identified several objectives. First I wanted the system to validate and clean the user's input before any planning took place. Secondly I wanted it to generate a proposed academic title, research aim, objectives and research questions. Thirdly, I designed the system to suggest literature search keywords and identify information that researchers should gather before beginning detailed investigation. And finally, I wanted the application to demonstrate good software engineering practice through modular architecture, documentation and automated testing.

So this slide illustrates how the researchmate intelligent agent operates. The workflow begins when a user enters a research topic. As you can see. The processes the processor module validates and cleans the input before passing it to the planner. I deliberately separated these stages because validation should always occur before planning and the planner then constructs the proposed academic title, research aim, objectives and research questions. And finally, the output generator formats everything into a structured academic report. One design decision I was particularly pleased with and proud of is where I separated each responsibility into its own module. This makes the system easier to sort of maintain, easier to test and much easier to

extend in the future. For example, I could replace the current deterministic planner with a large languaging model without redesigning the rest of the application.

So during implementation I deliberately followed recognised software engineering principles. The application begins in `main.py` which simply launches the application. The research mate agent coordinates the workflow while specialist processing is dedicated to separate modules. I chose this modular approach because I wanted every component to have a single responsibility. this would essentially you know improve readability, maintainability and automated testing, essentially. Another important design decision was using deterministic planning rather than immediately integrating a live languaging model. So although this reduced flexibility, it gave me predictable outputs, simplified debugging and allowed me to build reliable automated tests. My intention was to establish a stable software architecture first before introducing let's say more advanced AI capabilities.

So this slide is interesting. It demonstrates the working prototype of a researchmate in action. Apologies that the screenshots here are a bit small, but I'll explain. So the first screenshot shows the application running from the terminal. When the program starts it prompts the user to enter a research topic or research problem. In this example I entered the impact of generative AI and cyber security for example. Once the user submits the topic the agent begins executing its workflow. The first stage validates and cleans the input to ensure it is suitable for processing. Next the planner module generates a structured academic research plan on the topic. the retriever module then prepares literature search keywords and identifies the types of information the researcher would need to collect. Finally the output generator formats everything into a clean and structured report. And the second screenshot shows part of the generative, generated output rather than returning a simple paragraph of text, the research mate produces multiple academic planning components including a proposed research title, research aim, objectives research questions, methodology considerations, literature search keywords, information requirements and recommended planning steps and also a confidence note explaining that the result should be reviewed by a researcher.

And the final screenshot demonstrates the testing phase. I implemented both unit tests and functional workflow tests using `py.test`. The results show that all eight tests pass successfully providing evidence that the application's individual modules and overall workflow are functionally correct. It actually took a few tests for me to do this and I've provided some of those examples within the evidence folder and yeah so, essentially just to reiterate that the system can successfully receive a user goal produce it for a modular planning workflow generate meaningful structured outputs and operate reliably throughout user testing

Essentially from the beginning, I wanted testing to be an integral part of the development process rather than something completed at the end. I therefore implemented both unit tests and functional tests. Unit tests verify individual modules while functional tests will confirm that the complete intelligent agent workflow behaved correctly. And during the development I encountered several issues. For example, I experienced problems with uh test discovery, Python module recognition and input validation. So rather than working around these issues, I

corrected the underlying causes and then reran the automated test to verify that the fixes have been successful. As shown in the test results, all eight automated tests pass successfully. This gave me confidence that the current implementation behaves consistently and provides a reliable foundation for future development.

So looking back, I believe the greatest strength of research made is in its architecture essentially. I'm particularly pleased with the module design, the modular design, excuse me, because it makes the application significantly easier to maintain, extend, and test. However, I also recognise several limitations. The current version uses deterministic planning rather than a large language model. It doesn't retrieve academic literature automatically and currently operates through a command line interface. I also considered the ethical implications of AI assisted academic work. Researchmate is designed to support academic planning rather than replace critical thinking or academic supervision. Users should always review and validate the generated outputs but before relying on them in their own academic work.

So although the current implementation demonstrates the core planning workflow, I deliberately designed the architecture so that future capabilities can be added without redesigning the application. So my first priority would be integrating a live large language model to generate more context aware research planning. After that, retrieving augmented generation could allow the application to retrieve information directly from trusted academic databases. Additionally improvements could include citation validation, maybe methodology recommendations, document export and a web based interface, essentially. Because each responsibility already exists as an independent module in these improvements can be improved incrementally whilst preserving the existing architecture. So there's room for it to scale essentially.

And in conclusion, research successfully demonstrates the design and implementation of an intelligent academic research planning and decision support agent. The application receives a user's research goal. It validates the input, generates a structured research plan, and produces meaningful outputs for a modular workflow. And throughout the project I focus not only on building working software but on building software that is maintainable, testable and capable of future enhancement and growth. Looking back I'm pleased with the final solution that you know, pleased that the final solution demonstrates both intelligent agent concepts and sound engineer software engineering principles. Although the current implementation represents simply a proof of concept, I believe it establishes a strong foundation for future AI enabled academic research support. Overall I'm very happy that this is something I attempted to do and that I essentially got through it. So thank you very much for listening. I hope this presentation has demonstrated not only what I built but also the reasoning behind the design decisions that I made.

Thank you.